Supplementary information

Identifiability of spatial relations under geodetic datum and reference-frame loss

S1 Formal assumptions and degeneracies

The distance result assumes a labeled Euclidean distance matrix of embedding dimension at most three. Sparse graphs may admit flexes or multiple discrete realizations. Repeated points, collinear or coplanar simplices, axis-pole locations, and projected antipodes are treated as degeneracies rather than forced directional classifications.

Generic axial ordering requires an origin and a directed axis. An undirected axis leaves its labels exchangeable. Signed relative longitude requires a physically recoverable chiral witness; absolute angular origin requires an additional directed meridian. A stored orientation bit without recoverable physical meaning does not remove the reflection ambiguity.

S2 Computational algorithms

Classical multidimensional scaling uses the three largest positive eigenpairs of the double-centered squared-distance matrix. Procrustes alignment uses a singular-value decomposition and explicitly constrains the determinant when reflection is forbidden. Direction signs are evaluated from normalized axial projections and a two-argument arctangent on the principal branch. The rigidity analysis uses numerical rank of the three-dimensional rigidity matrix with an SVD tolerance scaled by matrix dimensions and machine precision.

S3 Reproducibility configuration


```
seed: 20260924
synthetic:
  n_points: 24
  n_frame_trials: 250
  monte_carlo_trials: 4000
  ellipsoid:
    name: WGS84
    semi_major_axis_m: 6378137.0
    inverse_flattening: 298.257223563
  ellipsoid_sensitivity:
    name: GRS80
    semi_major_axis_m: 6378137.0
    inverse_flattening: 298.257222101
  anchor_corruption_probabilities: [0.0, 0.01, 0.05, 0.1, 0.2, 0.4]
  angular_noise_degrees: [0.0, 0.01, 0.05, 0.1, 0.25, 0.5]
rigidity:
  n_points: [6, 8, 10, 12, 16]
  edge_probabilities: [0.1, 0.2, 0.3, 0.4, 0.5, 0.6, 0.7, 0.8, 0.9, 1.0]
  trials: 2000
  seed_offset: 1000
real_data:
  source: data/raw/itrf2020/ITRF2020_SLR.SSC.txt
  reference_epoch: 2015.0
  comparison_epoch: 2025.0
```

```
max_coordinate_sigma_m: 0.02
n_sites: 16
```

S4 ITRF2020 selected stations

Table S1 Selected ITRF2020 SLR stations and epochs.

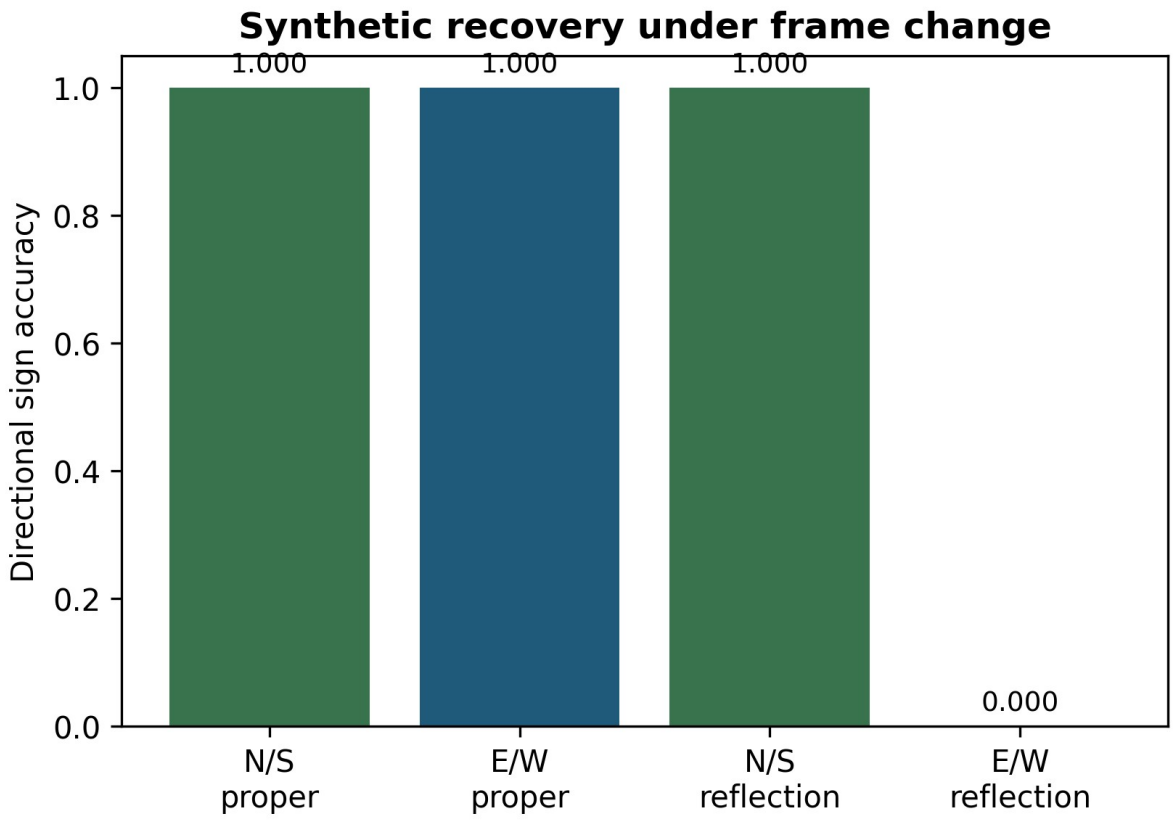
Domes	Site Name	Technique Id	Latitude Degrees	Longitude Degrees	Reference Epoch	Comparison Epoch
40445M004	MAUI	7119	20.579588011218565	-156.25692679418262	2015.0	2025.0
40497M001	Monument Peak	7110	32.716542819713	-116.4226784499638	2015.0	2025.0
40442M006	FORT DAVIS	7080	30.5117098111711	-104.01519959418388	2015.0	2025.0
40451M105	Washington	7105	38.83248993117698	-76.8276968652703	2015.0	2025.0
48081S001	Campus Universi	7407	-15.67269811883136	-47.86529338063169	2015.0	2025.0
13402S007	SAN FERNANDO	7824	36.28149367442857	-6.2053067139107 8	2015.0	2025.0
14001S007	Zimmerwald	7810	46.68520770658857	7.465224516900016	2015.0	2025.0
30302M003	Hartebeesthoek	7501	-25.73887838935569	27.686177286269704	2015.0	2025.0
25603S001	Baikonur Cosmod	1887	45.51259179864986	63.342416825669 55	2015.0	2025.0
12372S001	Altay	1879	51.15604735269942	82.177298365774 05	2015.0	2025.0
12338S004	Badary	1890	51.58283480322939	102.23536268348 586	2015.0	2025.0
50107M001	Yarragadee	7090	-28.883434983501672	115.346753302934	2015.0	2025.0
50119S003	Mount Stromlo	7825	-35.13482884193121	149.0098822886652	2015.0	2025.0
12734S008	Matera	7941	40.458577942210	16.704614547110	2015.0	2025.0

Domes	Site Name	Technique Id	Latitude Degrees	Longitude Degrees	Reference Epoch	Comparison Epoch
			4	5		
13212S001	Herstmonceux Ca	7840	50.67885528360768	0.3361285649584317	2015.0	2025.0
10002S002	Grasse (OCA)	7845	43.56246101585594	6.921576461282378	2015.0	2025.0

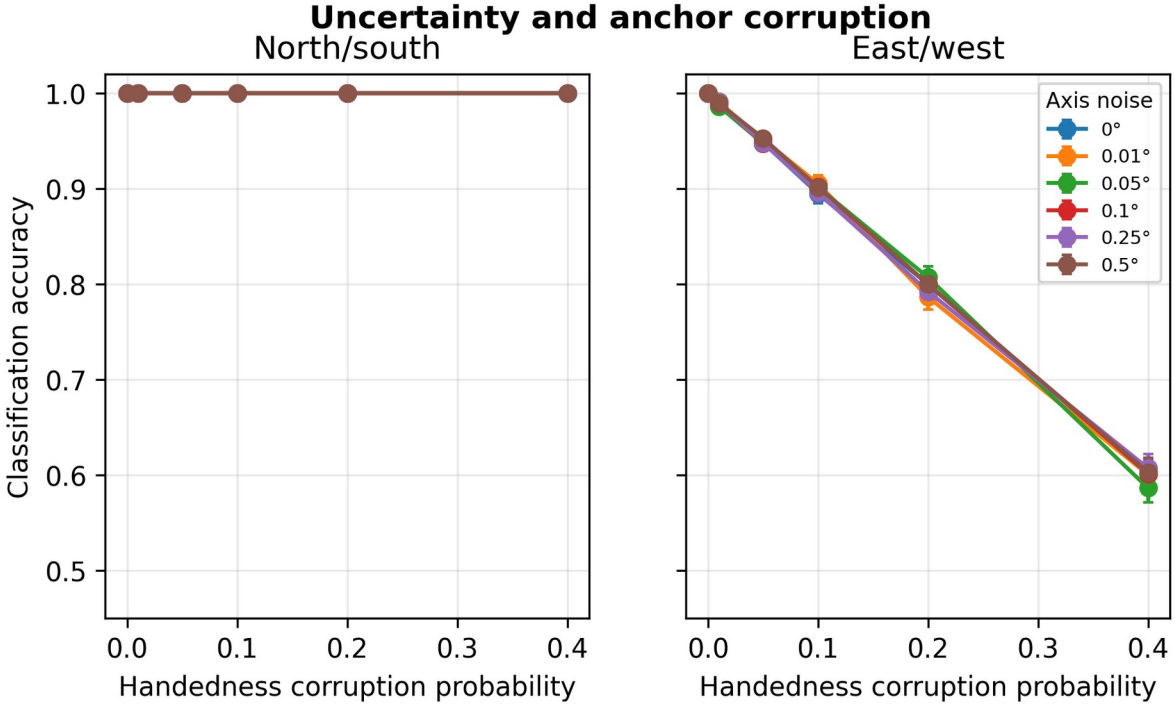
S5 Incomplete-network and ellipsoid sensitivity

Table S2 WGS84–GRS80 sensitivity for the seeded point set.

Metric	Value	Unit
ellipsoid_maximum_coordinate_difference_m	0.0001210656051966	m
ellipsoid_median_coordinate_difference_m	0.0001025915363387	m
ellipsoid_maximum_chord_distance_difference_m	0.0002250224351882	m
ellipsoid_median_chord_distance_difference_m	4.020938649773598e-05	m
ellipsoid_axial_ordering_agreement	1.0	proportion
ellipsoid_signed_longitude_agreement	1.0	proportion



Supplementary Fig. S1 Synthetic theorem-validation benchmark. Proper rotations preserve both tested directional signs; axial reflection preserves axial ordering and reverses signed longitude.



Supplementary Fig. S2 Seeded axis-noise and algebraic-orientation-corruption experiment. The curves are conditional simulation checks and are not estimates of physical anchor survival.

S6 Numerical result registry

Table S3 Machine-generated values used in the manuscript.

Metric	Value	Source	Unit
synthetic_number_of_points	24.0	synthetic_benchmark	proportion_or_m
synthetic_frame_trials	250.0	synthetic_benchmark	proportion_or_m
synthetic_monte_carlo_trials_per_condition	4000.0	synthetic_benchmark	proportion_or_m
distance_reconstruction_rmse_allow_reflection_m	2.970170222267324e-09	synthetic_benchmark	proportion_or_m
distance_reconstruction_rmse_proper_only_m	6136173.251345326	synthetic_benchmark	proportion_or_m
proper_rotation_north_south_accuracy	1.0	synthetic_benchmark	proportion_or_m
proper_rotation_east_west_accuracy	1.0	synthetic_benchmark	proportion_or_m
reflection_north_south_accuracy	1.0	synthetic_benchmark	proportion_or_m
reflection_east_west_accuracy	0.0	synthetic_benchmark	proportion_or_m
n_selected_sites	16.0	itrf2020_demonstration	count
maximum_coordinate_standard_uncertainty_m	0.02	itrf2020_demonstration	m
distance_reconstruction_rmse_m	3.613275939091764e-09	itrf2020_demonstration	m
axial_reflection_north_south_accuracy	1.0	itrf2020_demonstration	proportion
axial_reflection_east_west_accuracy	0.0	itrf2020_demonstration	proportion
maximum_pairwise_distance_change_m	0.6906252605840564	itrf2020_demonstration	m
median_pairwise_distance_change_m	0.0907241012901067	itrf2020_demonstration	m
reference_epoch	2015.0	itrf2020_demonstration	decimal_year
comparison_epoch	2025.0	itrf2020_demonstration	decimal_year
random_seed	20260924.0	configuration	integer
ellipsoid_maximum_coordinate_difference_m	0.0001210656051966	ellipsoid_sensitivity	m
ellipsoid_median_coordinate_difference_m	0.0001025915363387	ellipsoid_sensitivity	m
ellipsoid_maximum_chord_distance_difference_m	0.0002250224351882	ellipsoid_sensitivity	m
ellipsoid_median_chord_distance_difference_m	4.020938649773598e-05	ellipsoid_sensitivity	m

Metric	Value	Source	Unit
ellipsoid_axial_ordering_agreement	1.0	ellipsoid_sensitivity	proportion
ellipsoid_signed_longitude_agreement	1.0	ellipsoid_sensitivity	proportion
rigidity_largest_network_n_points	16.0	rigidity_phase_diagram	count
rigidity_trials_per_cell	2000.0	rigidity_phase_diagram	count
rigidity_95pct_threshold_edge_probability	0.6	rigidity_phase_diagram	probability
rigidity_probability_at_95pct_threshold	0.996	rigidity_phase_diagram	proportion
rigidity_95pct_threshold_95ci_lower	0.992126535703962	rigidity_phase_diagram	proportion
rigidity_95pct_threshold_95ci_upper	0.9979717533930348	rigidity_phase_diagram	proportion
rigidity_edge_probability_before_95pct_threshold	0.5	rigidity_phase_diagram	probability
rigidity_probability_before_95pct_threshold	0.9465	rigidity_phase_diagram	proportion
rigidity_probability_before_95pct_threshold_95ci_lower	0.935754264314357	rigidity_phase_diagram	proportion
rigidity_probability_before_95pct_threshold_95ci_upper	0.955533812463484	rigidity_phase_diagram	proportion
north_south_accuracy_at_0.1_chirality_corruption	1.0	uncertainty_benchmark	proportion
east_west_accuracy_at_0.1_chirality_corruption	0.895	uncertainty_benchmark	proportion
east_west_accuracy_at_0.1_chirality_corruption_95ci_lower	0.8851180041126476	uncertainty_benchmark	proportion
east_west_accuracy_at_0.1_chirality_corruption_95ci_upper	0.9041240356884884	uncertainty_benchmark	proportion
north_south_accuracy_at_0.4_chirality_corruption	1.0	uncertainty_benchmark	proportion
east_west_accuracy_at_0.4_chirality_corruption	0.60225	uncertainty_benchmark	proportion
east_west_accuracy_at_0.4_chirality_corruption_95ci_lower	0.5869914415518415	uncertainty_benchmark	proportion
east_west_accuracy_at_0.4_chirality_corruption_95ci_upper	0.6173123522954146	uncertainty_benchmark	proportion

Spatial relations under datum loss

S7 Test coverage

The automated suite covers rigid-transform distance invariance, independent proper and reflection-permitting alignment, simplex chirality, local ENU basis orthonormality, proper-rotation direction invariance, axial-reflection direction asymmetry, branch wrapping, same-meridian ties, pole and exact/near-antipode degeneracies, rigidity rank, ITRF parsing and site count, configuration-dependent epoch labels, nondefault uncertainty grids, and seeded reproducibility.

S8 Interpretation of main numerical values

The proper-only mirror ambiguity is 6.136 million m for the seeded point set. This value characterizes the selected configuration and is not a universal Earth-scale constant.

The local-rigidity frequencies are Monte Carlo summaries for independently sampled generic points and Bernoulli edge retention. They do not establish global rigidity, geodetic network optimality, or a universal density threshold.